MIL-M-28787/242A(EC)
14 May 1982
SUPERSEDING
MIL-M-28787/242(EC)
24 January 1980

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, INTERFACE, NTDS(ANEW), DIGITAL, KEY CODE QDF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the cquisition of an electronic NTDS(ANEW) interface receiver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the ollowing specifications and standards, of the issue listed in that issue of the epartment of Defense Index of Specifications and Standards specified in the olicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.198 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be QDF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/242-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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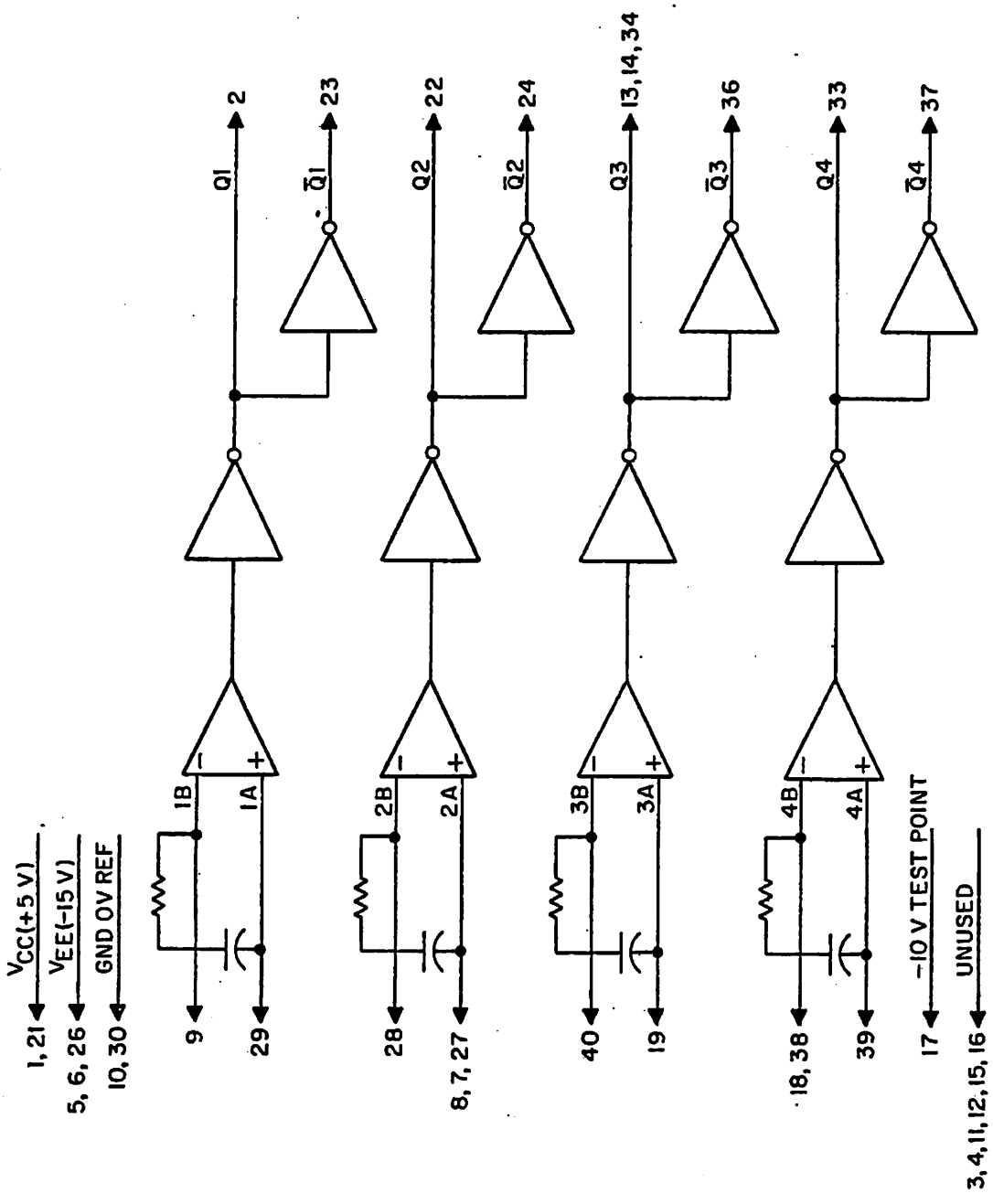


*FIGURE 1. QDF module function diagram NTDS (ANEW) interface receiver.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	-	10	-	10	-	10	-	MΩ
B1-B11	Isolation	III	X	10	-	10	-	10	-	10	-	MΩ
C1-C9	Continuity	IV	X	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
D1-D8	V_{OUT} (low-loaded)	V	S	0.1	0.8	0.1	0.8	0.1	0.8	0.1	0.8	V
E1-E4	Output short circuit current, Q outputs	VI	X	-1.4	-4.0	-1.4	-4.0	-1.4	-4.0	-1.4	04.0	mA
E5-E8	Output short circuit current, $\overline{Q}$ outputs	VI	X	-20	-55	-20	-55	-20	-55	-20	-55	mA
F1-F8	Static output voltage, low level, Q and $\overline{Q}$ outputs	VII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
G1-G4	Static output voltage, high level, Q outputs	VIII	X	2.8	4.5	2.8	4.5	2.8	4.5	2.8	4.5	V
G5-G8	Static output voltage, high level, $\overline{Q}$ outputs	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H4	Inputs open/outputs low	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J8	Input terminal impedance check	X	S	1.6	2.4	1.6	2.4	1.6	2.4	1.6	2.4	V
K1-K4	Input resistance balance check	XI	S	0.92	1.08	0.92	1.08	0.92	1.08	0.92	1.08	Ratio
L1-L8	Input current (V_{IN} = +10 V), differential inputs	XII	X	0.1	3.5	0.1	3.5	0.1	3.5	0.1	3.5	mA
M1-M8	Input current (V_{IN} = -10 V), differential inputs	XII	X	-1.0	-7.0	-1.0	-7.0	-1.0	-7.0	-1.0	-7.0	mA
N1-N4	Input current (V_{IN} = +3 V)	XIII	X	-1.0	2.5	-1.0	2.5	-1.0	2.5	-1.0	2.5	mA
N5-N8	Input current (V_{IN} = 0 V)	XIII	X	-0.1	-2.5	-0.1	-2.5	-0.1	-2.5	-0.1	-2.5	mA
P1	Power supply current	XIV	X	10	49	10	49	10	49	10	49	mA
P2	Power supply current	XIV	X	10	38	10	38	10	38	10	38	mA
Q1-Q2	Power supply current	XIV	X	-15	-65	-15	-65	-15	-65	-15	-65	mA
R1-R8	Transition time (t_{TLH}), Q outputs, C_L = 50 pF	XV	S	10	300	10	300	10	320	10	320	ns
R9-R16	Transition time (t_{TLH}), $\overline{Q}$ outputs, C_L = 50 pF	XV	S	5	35	5	35	5	40	5	40	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R17-R24	Transition time (t_{THL}), Q outputs, C_L = 50 pF	XV	S	5	30	5	30	5	35	5	35	ns
R25-R32	Transition time (t_{THL}), $\overline{Q}$ outputs, C_L = 50 pF	XV	S	5	25	5	25	5	30	5	30	ns
R33-R40	Transition time (t_{TLH}), Q outputs, C_L = 150 pF	XV	Q	10	310	10	310	10	330	10	330	ns
R41-R48	Transition time (t_{TLH}), $\overline{Q}$ outputs, C_L = 150 pF	XV	Q	5	40	5	40	5	45	5	45	ns
R49-R56	Transition time (t_{THL}), Q outputs, C_L = 150 pF	XV	Q	5	35	5	35	5	40	5	40	ns
R57-R64	Transition time (t_{THL}), $\overline{Q}$ outputs, C_L = 150 pF	XV	Q	5	30	5	30	5	35	5	35	ns
S1-S8, S17-S24	Propagation delay time (t_{PLH} and t_{PHL}), Q outputs, C_L = 50 pF	XVI	S	5	52	5	52	5	60	5	60	ns
S9-S16, S25-S32	Propagation delay time (t_{PLH} and t_{PHL}), $\overline{Q}$ outputs, C_L = 50 pF	XVI	S	8	76	8	76	8	81	8	81	ns
S33-S40, S49-S56	Propagation delay time (t_{PLH} and t_{PHL}), Q outputs, C_L = 150 pF	XVI	Q	5	58	5	58	5	66	5	66	ns
S41-S48, S57-S64	Propagation delay time (t_{PLH} and t_{PHL}), $\overline{Q}$ outputs, C_L = 150 pF	XVI	Q	8	82	8	82	8	87	8	87	ns
T1-T16	Common mode rejection check, low level, Q and $\overline{Q}$ outputs	XVII	S	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
U1-U4, U9-U12	Common mode rejection check, high level, Q outputs	XVII	S	2.8	4.5	2.8	4.5	2.8	4.5	2.8	4.5	V
U5-U8, U13-U16	Common mode rejection check, high level, $\overline{Q}$ outputs	XVII	S	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
V1-V8	Output crosstalk, low level	XVIII	Q	0	400	0	400	0	400	0	400	mV
W1-W8	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
X	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	mV
Y	Power supply transient rise time	XXI	S	3.0	8.5	3.0	8.5	3.0	8.5	3.0	8.5	µs
Z1-Z4	Input capacitance	XXII	Q	5	35	5	35	5	35	5	35	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through Y shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Y, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through Z shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage	
V_{CC}	5.0 V ±10%
V_{EE}	-15.0 V ±10%
Maximum power dissipation	1,342 mW (outputs not loaded)

Input voltage differential for positive logic output.

Low output ($\overline{Q}$)	-10.0 V min to 0.8 V max
Low output (Q)	2.2 V min to 10.0 V max
High output ($\overline{Q}$)	2.2 V min to 10.0 V max
High output (Q)	-10.0 V min to 0.8 max

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Input current	
Input low (source) (V_{IN} = 0.0 V)	-2.5 mA maximum
Input high (load) (V_{IN} = 3.0 V)	2.5 mA maximum
Output voltage	
High level (I_{OH} = 0.16 mA) (Q output)	2.8 V min
(I_{OH} = 0.4 mA) ($\overline{Q}$ output)	2.4 V min
Low level (I_{OL} = -10.8 mA) (Q output)	0.4 V max
(I_{OL} = -16 mA) ($\overline{Q}$ output)	0.4 V max
Output drive current	
High level (V_{OH} = 2.8 V) (Q output)	0.16 mA max
(V_{OH} = 2.4 V) ($\overline{Q}$ output)	0.4 mA max
Low level (V_{OL} = 0.4 V) (Q output)	-10.8 mA max
(V_{OL} = 0.4 V) ($\overline{Q}$ output)	-16.0 mA max

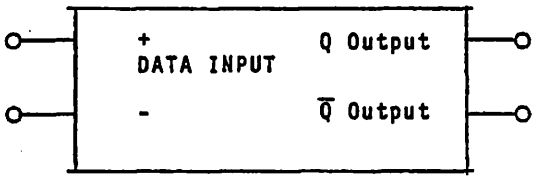
Propagation delay time	C_L = 50 pF	C_L = 150 pF
t_{PLH} (Q output)	5 ns min, 60 ns max	5 ns min, 66 ns max
t_{PLH} ($\overline{Q}$ output)	8 ns min, 81 ns max	8 ns min, 87 ns max
t_{PHL} (Q output)	5 ns min, 60 ns max	5 ns min, 66 ns max
t_{PHL} ($\overline{Q}$ output)	8 ns min, 81 ns max	8 ns min, 87 ns max

* 6.4 Module function. This module contains four independent interface line receivers that develop standard TTL positive logic output for line inputs that comply with NTDS(ANEW) interface requirements (MIL-STD-1397). Each of the four receivers provide both inverting and noninverting outputs.

6.4.1 Input to output relationship. Each interface receiver has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/-5/

INPUTS	OUTPUTS	
Differential	Q	$\overline{Q}$
H	L	H
L	H	L



1/ Differential input voltage is referred from plus input to minus input.
2/ Input "L" differential is - 10.0 V to 0.8 V.
3/ Input "H" differential is 2.2 V to 10.0 V.
4/ H = high level.
5/ L = low level.

6.4.2 Parallel outputs. Paralleling of output circuits for the wired OR operation is permissible for the Q outputs. Paralleling of the $\overline{Q}$ outputs is not permissible.

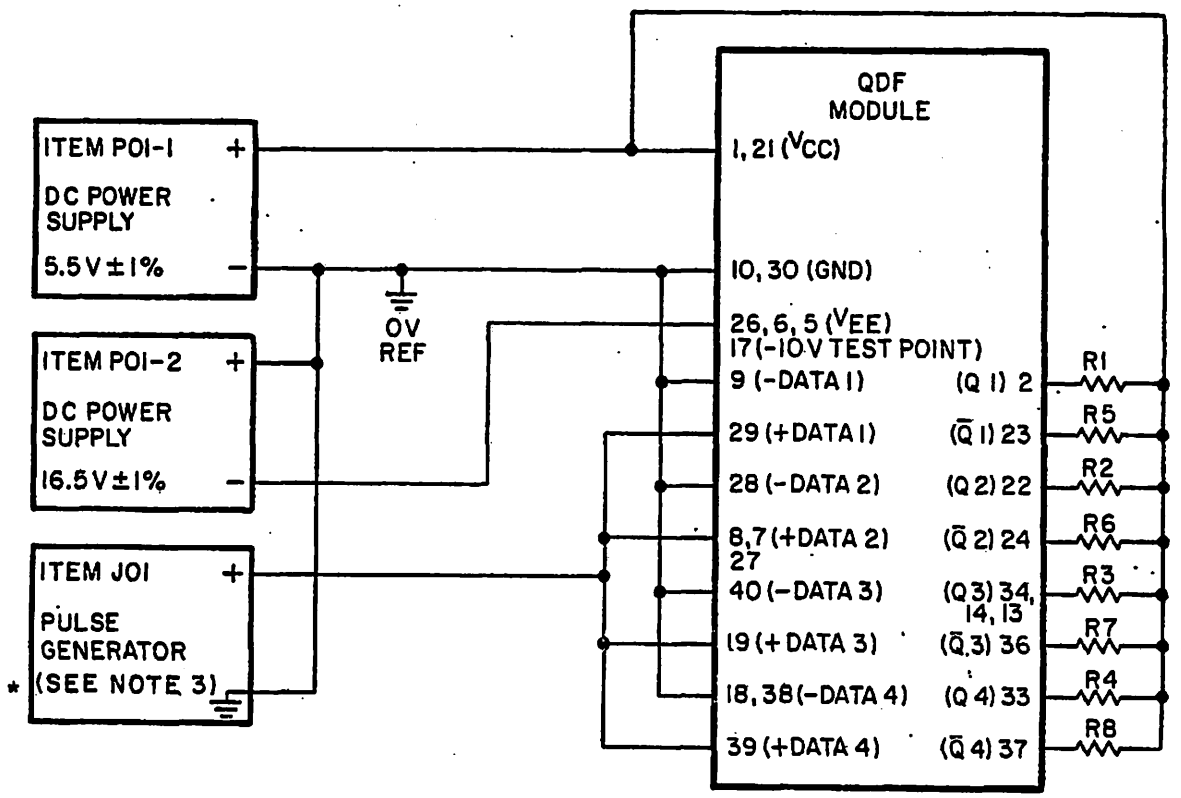
6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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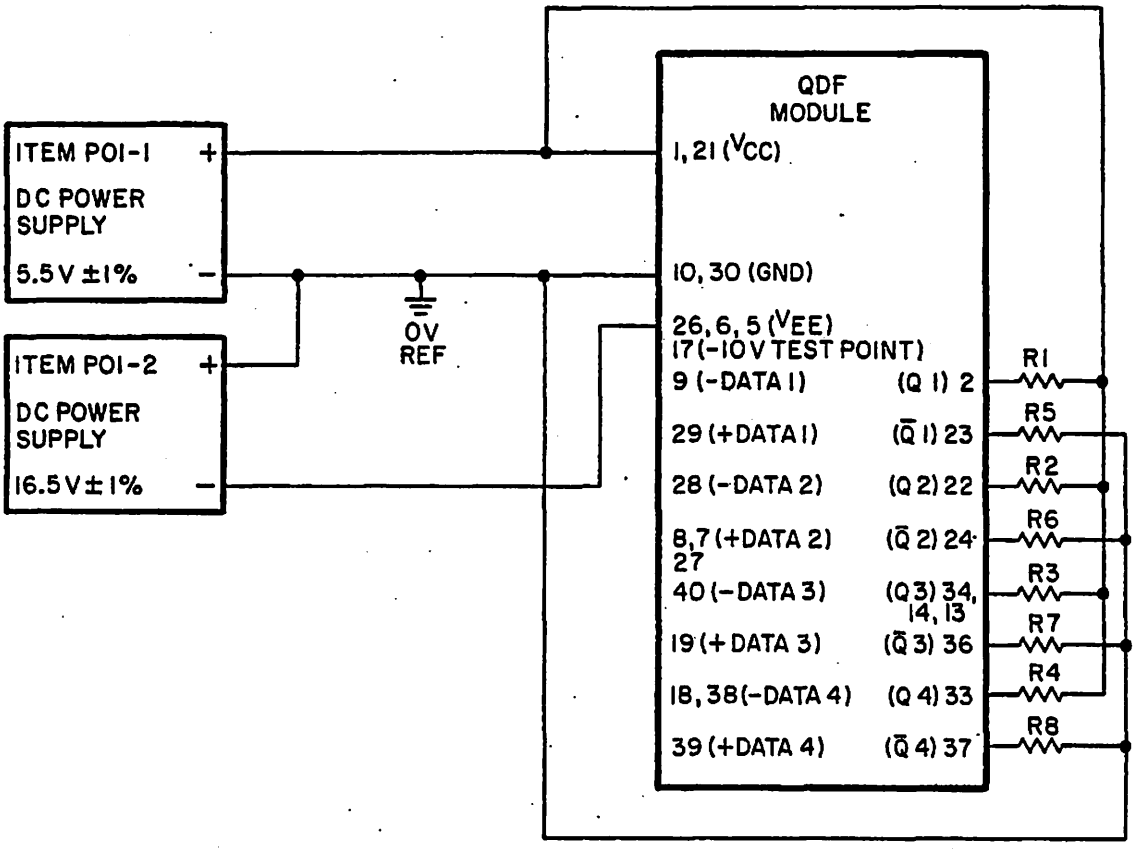
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NOTES:
1. R1-R4 = 511Ω ±1%, 1/4 W.
2. R5-R8 = 340Ω ±1%, 1/4 W.
3. Item JO1 setup:
   PRR = 1.0 MHz ±10%.
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 50 ±20 ns.
   PA = 2.3 ±0.1 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1-R4 = 511Ω ±1%, 1/4 W.
2. R5-R8 = 5.97 kΩ ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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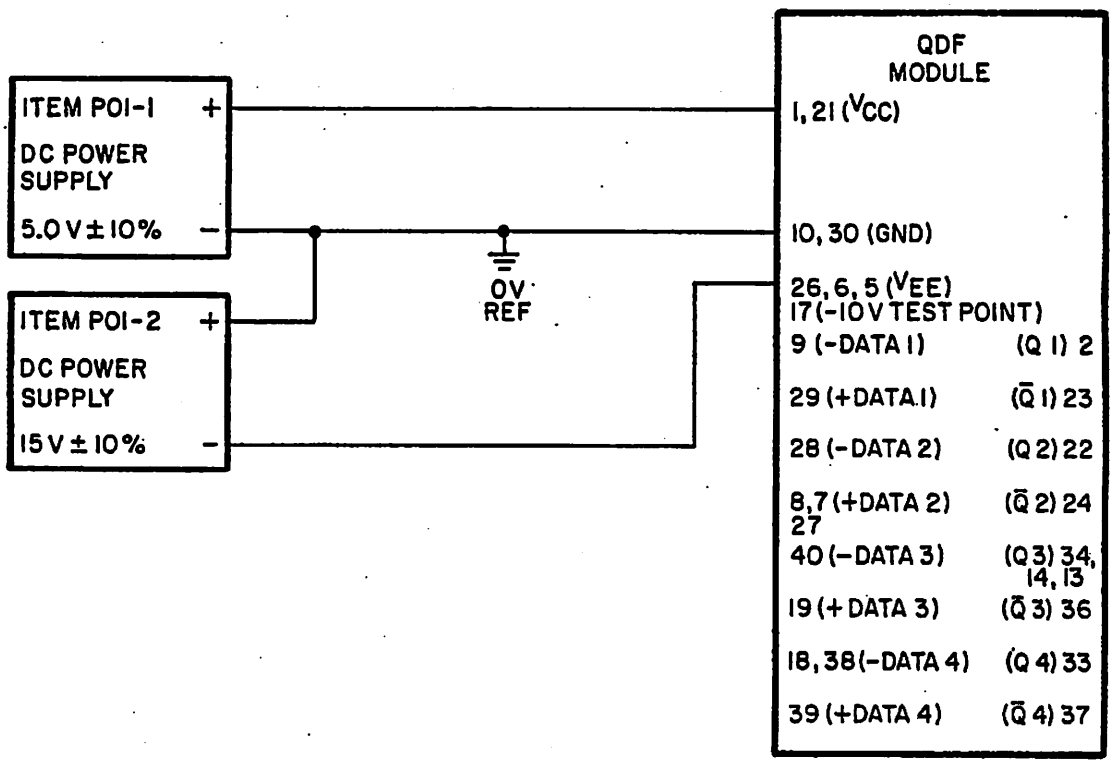


FIGURE 4. Vibration test circuit.

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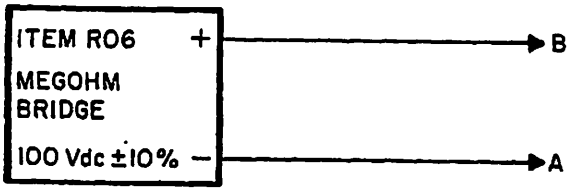
*TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	Frame	All other pins	10	∞	MΩ	5
B1		3					
B2		4					
B3		11					
B4		12					
B5		15					
B6		16					
B7		20					
B8		25					
B9		31					
B10		32					
B11		35					

TABLE IV. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Continuity	10	30	0.0	0.1	Ω	6
C2		26	6				
C3		26	5				
C4		8	7				
C5		8	27				
C6		18	38				
C7		34	14				
C8		34	13				
C9		1	21				

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QDF MODULE	
1, 21 (V_{CC})	
10, 30 (GND)	
26, 6, 5 (V_{EE})	
17 (-10V TEST POINT)	
9 (-DATA 1)	(Q 1) 2
29 (+DATA 1)	($\bar{Q}$ 1) 23
28 (-DATA 2)	(Q 2) 22
8, 7 (+DATA 2)	($\bar{Q}$ 2) 24
27	
40 (-DATA 3)	(Q 3) 34, 14, 13
19 (+DATA 3)	($\bar{Q}$ 3) 36
18, 38 (-DATA 4)	(Q 4) 33
39 (+DATA 4)	($\bar{Q}$ 4) 37

WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A, penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 5. Isolation test circuit for table III.

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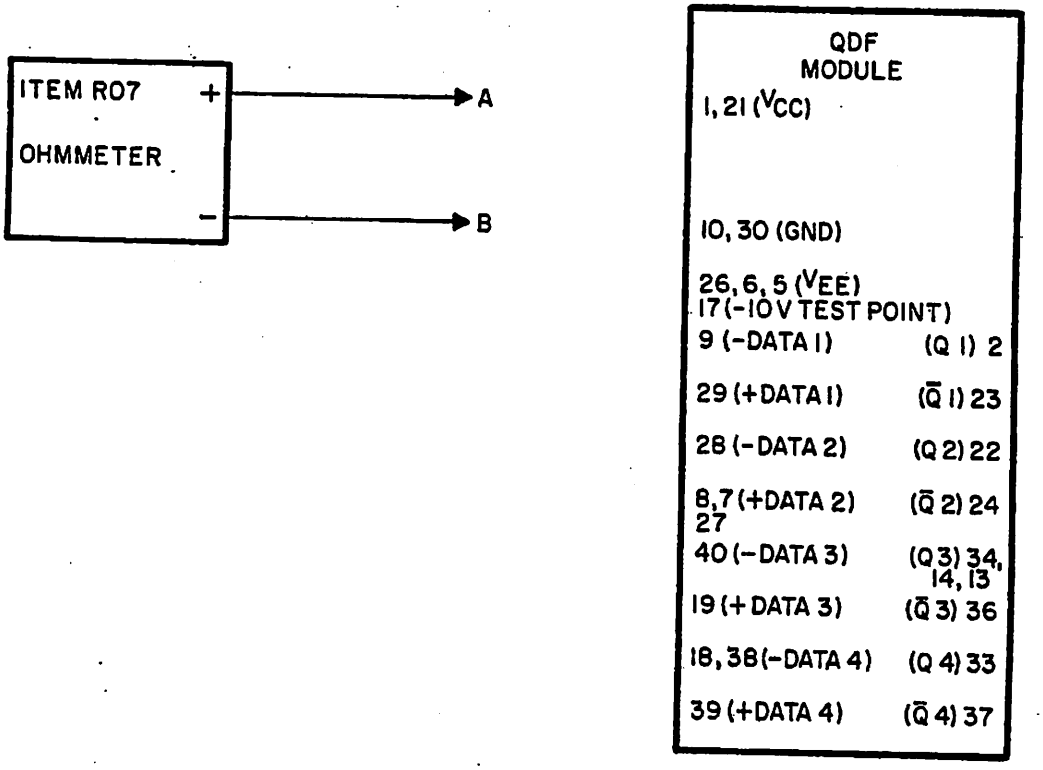


FIGURE 6. Continuity test circuit for table IV.

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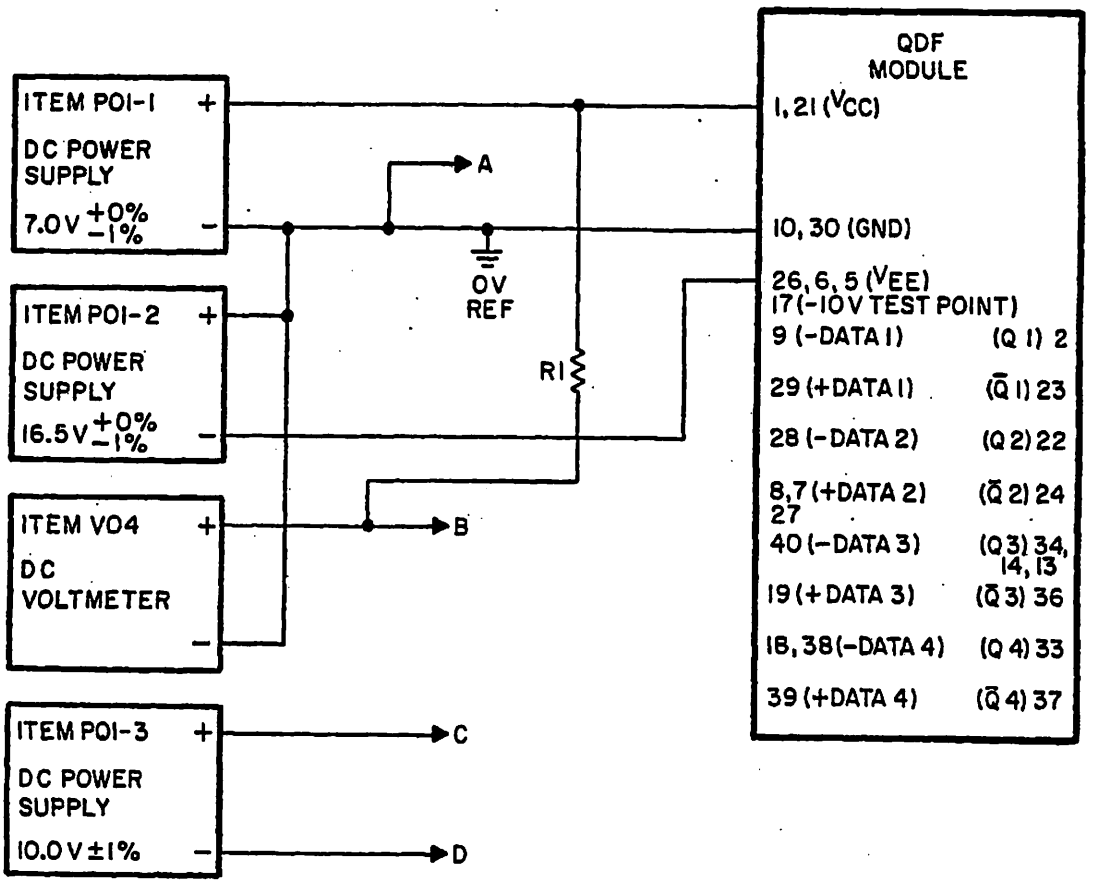
TABLE V. V_{OUT} (low-loaded) test

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	Q1 V_{OUT} (low-loaded)	9,28,40, 18,38	2	29,8,7,27, 19,39	10	0.1	0.8	V	7
D2	Q2		22						
D3	Q3		34,14,13						
D4	Q4		33						
D5	$\overline{Q}$		23	10	29,8,7, 27,19,39				
D6	$\overline{Q}2$		24						
D7	$\overline{Q}3$		36						
D8	$\overline{Q}4$		37						

TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	Q1 Output short circuit current	2	9	29	-1.4	-4.0	mA	8
E2	Q2	22	28	8,7,27				
E3	Q3	34,14,13	40	19				
E4	Q4	33	18,38	39				
E5	$\overline{Q}1$	23	29	9	-20	-55		
E6	$\overline{Q}2$	24	8,7,27	28				
E7	$\overline{Q}3$	36	19	40				
E8	$\overline{Q}4$	37	39	18,38				

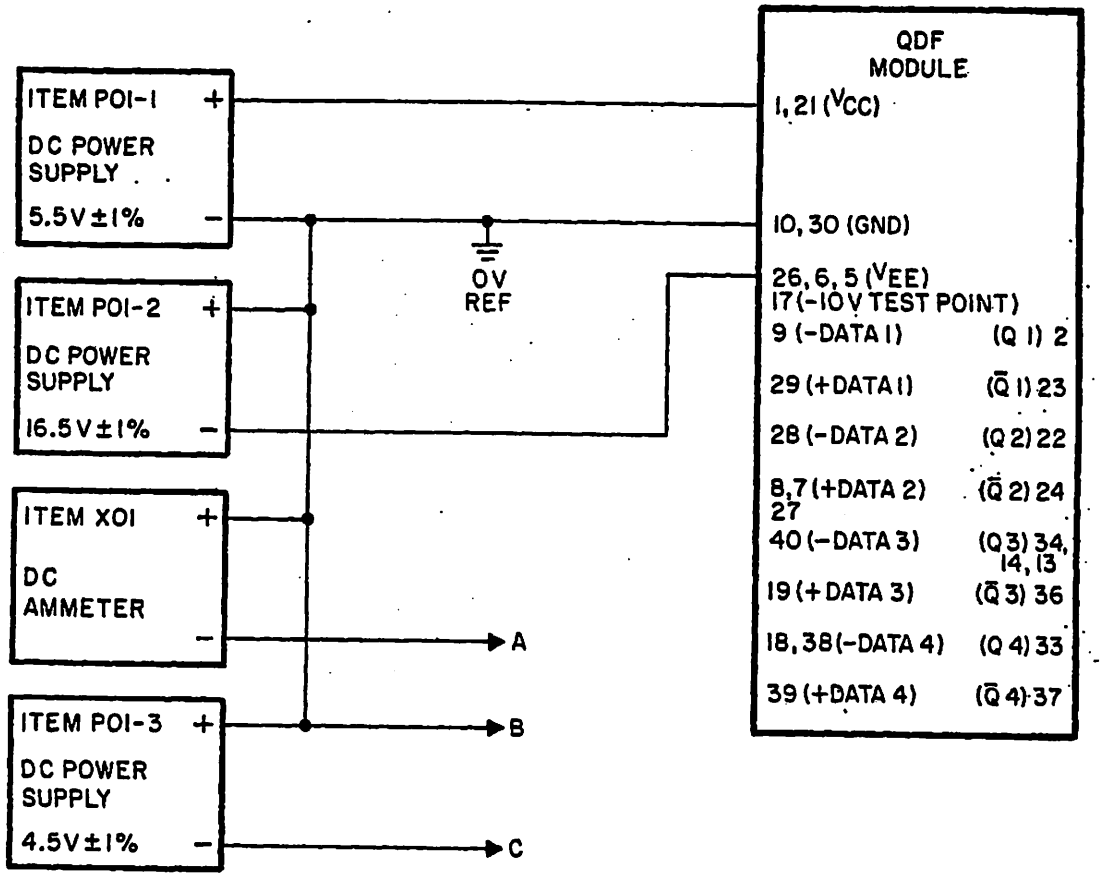
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NOTES:
1. RI = 383Ω ±1%, 1/4 W for test conditions D1-D4.
   RI = 255Ω ±1%, 1/4 W for test conditions D5-D8.
2. Each output shall be monitored for a minimum of 5 seconds before the measurement is taken.
3. If tests are successively repeated, a time interval of 10 seconds minimum between tests shall be allowed to prevent module overheating.

FIGURE 7. V_{OUT} (low-loaded) test circuit for table V.

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CAUTION:
Do not isolate the chassis of ammeter XOI from earth ground.

NOTE: When performing short circuit current tests, only one output shall be shorted at a time, and measurement taken within 5 seconds.

FIGURE 8. Output short circuit current test circuit for table VI.

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*TABLE VII. Static output voltage (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	Q1 Static output voltage, low level	9,28,40, 18,38	Open	29,8,7,27, 19,39	2	0.0	0.4	V	9
F2	Q2				22				
F3	Q3				34,14,13				
F4	Q4				33				
F5	$\overline{Q1}$		29,8,7,27, 19,39	Open	23				
F6	$\overline{Q2}$				24				
F7	$\overline{Q3}$				36				
F8	$\overline{Q4}$				37				

*TABLE VIII. Static output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	Q1 Static output voltage, high level	9,28,40, 18,38	29,8,7,27, 19,39	Open	2	2.8	4.5	V	9
G2	Q2				22				
G3	Q3				34,14,13				
G4	Q4				33				
G5	$\overline{Q1}$		Open	29,8,7,27, 19,39	23	2.4			
G6	$\overline{Q2}$				24				
G7	$\overline{Q3}$				36				
G8	$\overline{Q4}$				37				

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*TABLE IX. Inputs open/outputs low test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	Inputs open/outputs low	Open	Open	Open	2	0.0	0.4	V	9
H2	↓	↓	↓	↓	22	↓	↓	↓	↓
H3					34,14,13				
H4					33				

TABLE X. Input terminal impedance check test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Input terminal impedance check	9	29	1.6	2.4	V	11
J2	↓	28	8,7,27	↓	↓	↓	↓
J3		40	19				
J4		18,38	39				
J5		9	29				
J6		28	8,7,27				
J7		40	19				
J8		18,38	39				

TABLE XI. Input resistance balance check test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
		Obtain ratio					
K1	Input resistance balance check	9,B	29,B	0.92	1.08	Ratio	12
K2	↓	28,B	8,7,27,B	↓	↓	↓	↓
K3		40,B	19,B				
K4		18,38,B	39,B				

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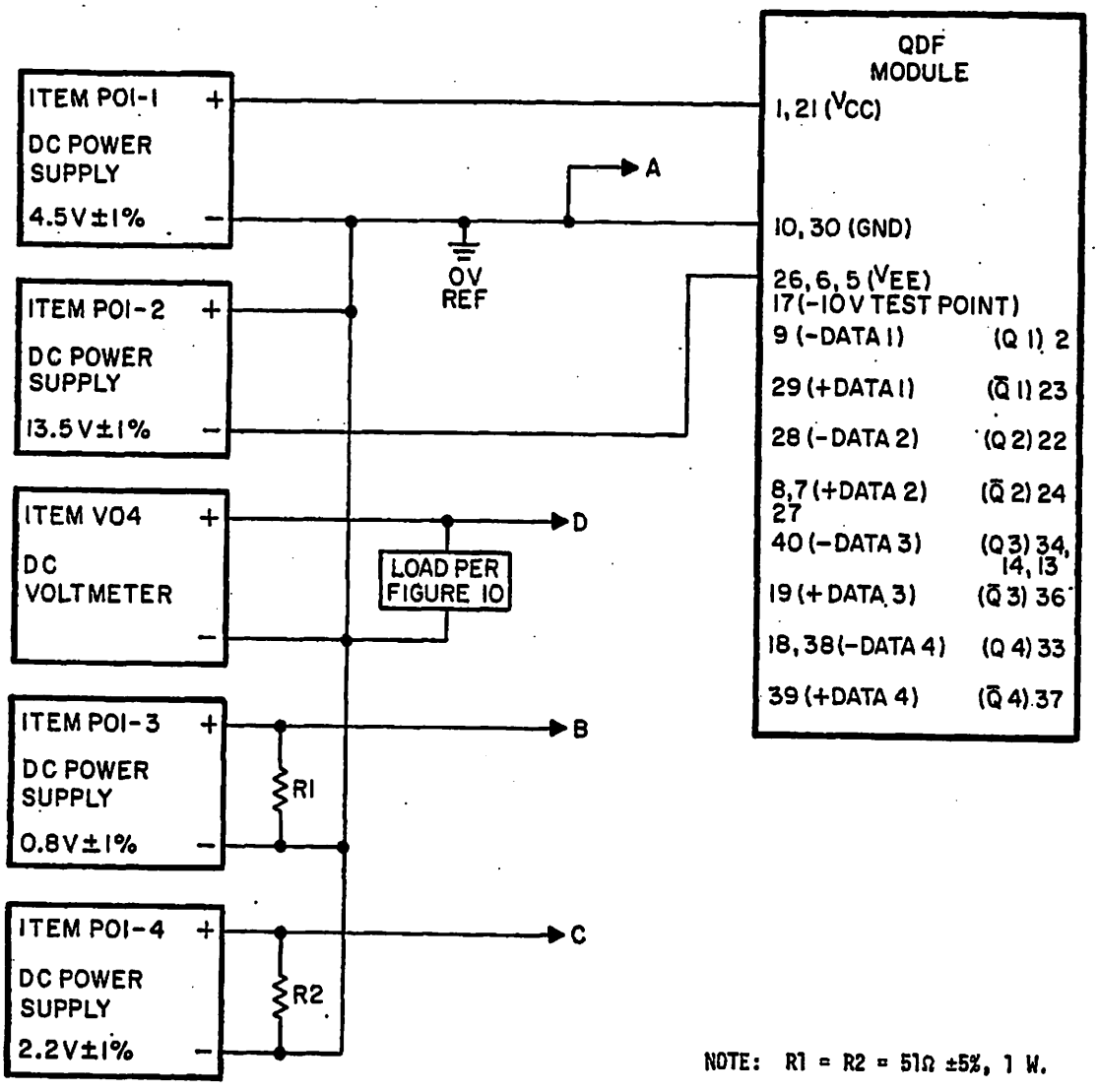
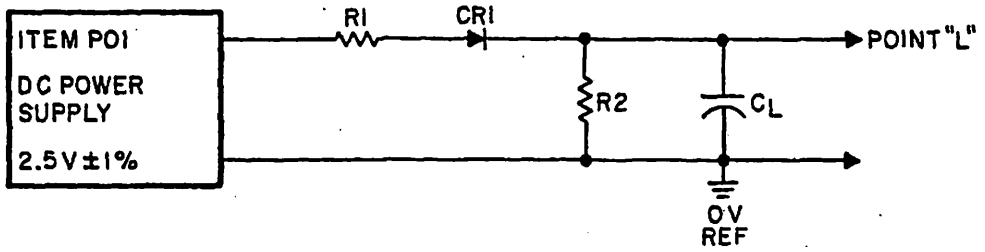


FIGURE 9. Static output voltage (low-level and high-level) and inputs open/outputs low test circuit for tables VII, VIII, and IX.

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NOTES:
Load circuit A - for Q outputs:

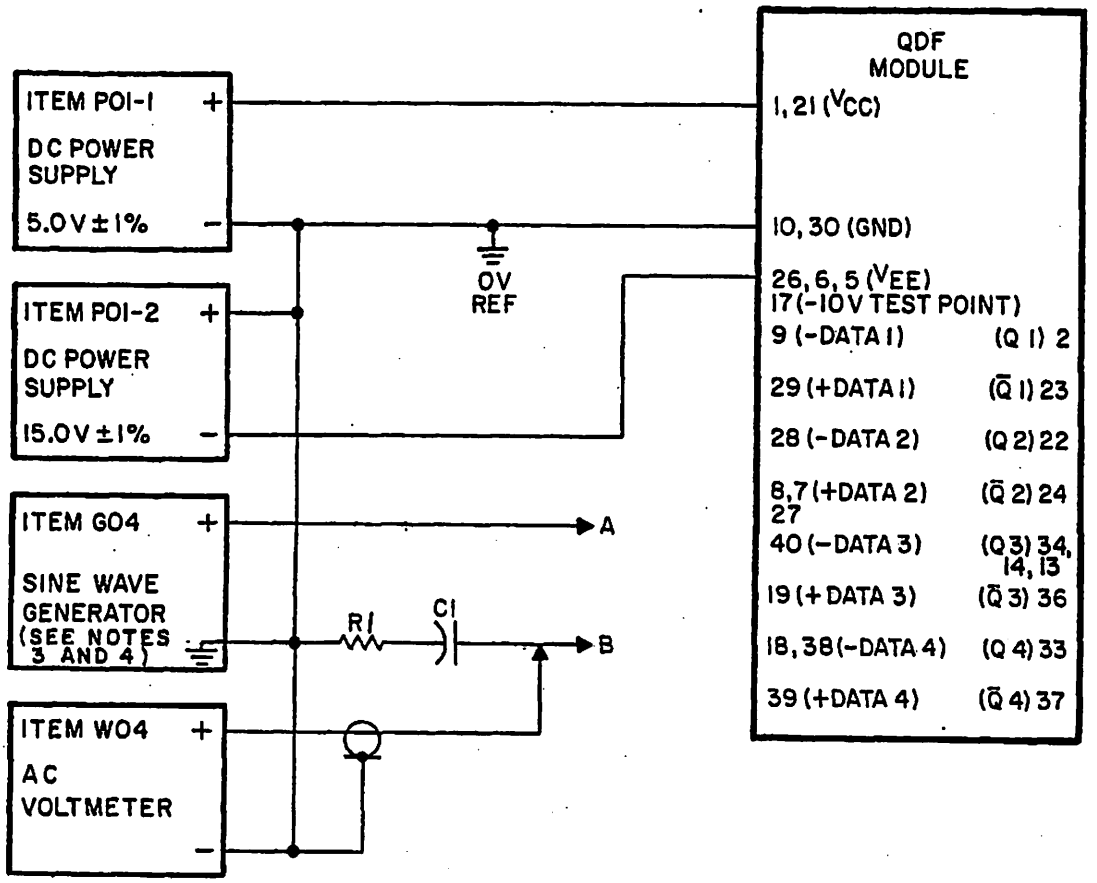
1. CR1 = 1N4150, or equivalent.
2. R1 = Select value for current flow of 10.8 mA ±5% out of point L with point L held at 0.4 V.
3. R2 = 17.8 kΩ ±1%, 1/8 W.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions R1-R8, R17-R24, S1-S8, S17-S24, T1-T4, U1-U4, V1-V8(odd), and W1-W8(odd).
5. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions R33-R40, R49-R56, S33-S40, S49-S56, T9-T12, and U9-U12.

Load circuit B - for $\overline{Q}$ outputs:

1. CR1 = 1N4150, or equivalent.
2. R1 = Select value for current flow of 16.0 mA ±5% out of point L with point L held held at 0.4 V.
3. R2 = 6.04 kΩ ±1%, 1/8 W.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions R9-R16, R25-R32, S9-S16, S25-S32, T5-T8, U5-U8, V1-V8(even), and W1-W8(even).
5. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions R41-R48, R57-R64, S41-S48, S57-S64, T13-T16, and U13-U16.

FIGURE 10. QDF logic load circuit.

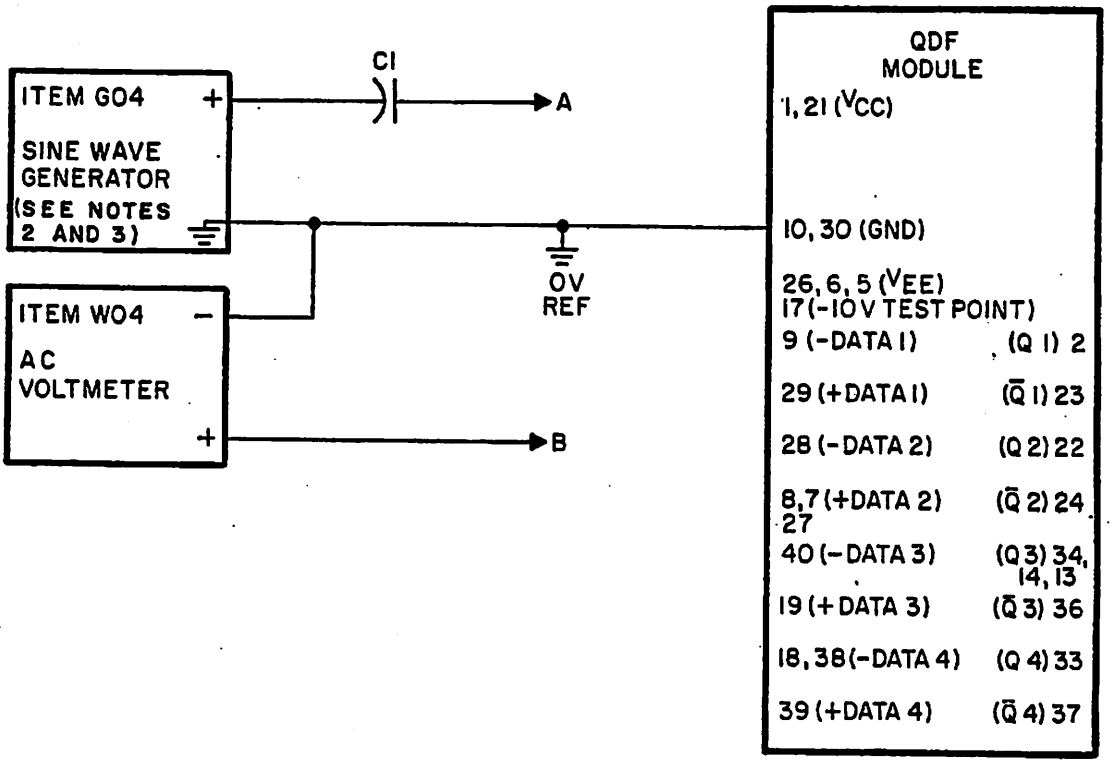
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NOTES:
1. R1 = 133Ω ±1%, 1/4 W.
2. C1 = 8200 pF ±1%.
3. For resistive component measurements test conditions J1-J4, set item G04 to 1.0 MHz ±5% and 4.0 Vrms ±1% measured at point A.
4. For capacitive component measurements test conditions J5-J8, set item G04 to 10.0 kHz ±5% and 4.0 Vrms ±1% measured at point A.

* FIGURE 11. Input terminal impedance check test circuit for table X.

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NOTES:
1. C1 = 0.01 µF ±20%.
2. Adjust frequency of item G04 to 1.0 MHz ±5%.
3. Adjust the amplitude of item G04 to 1.0 Vrms ±1% at the generator side of capacitor C1, with point A connected.

* FIGURE 12. Input resistance balance check test circuit for table XI.

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TABLE XII. Input current (+10 V and -10 V differential inputs) test.

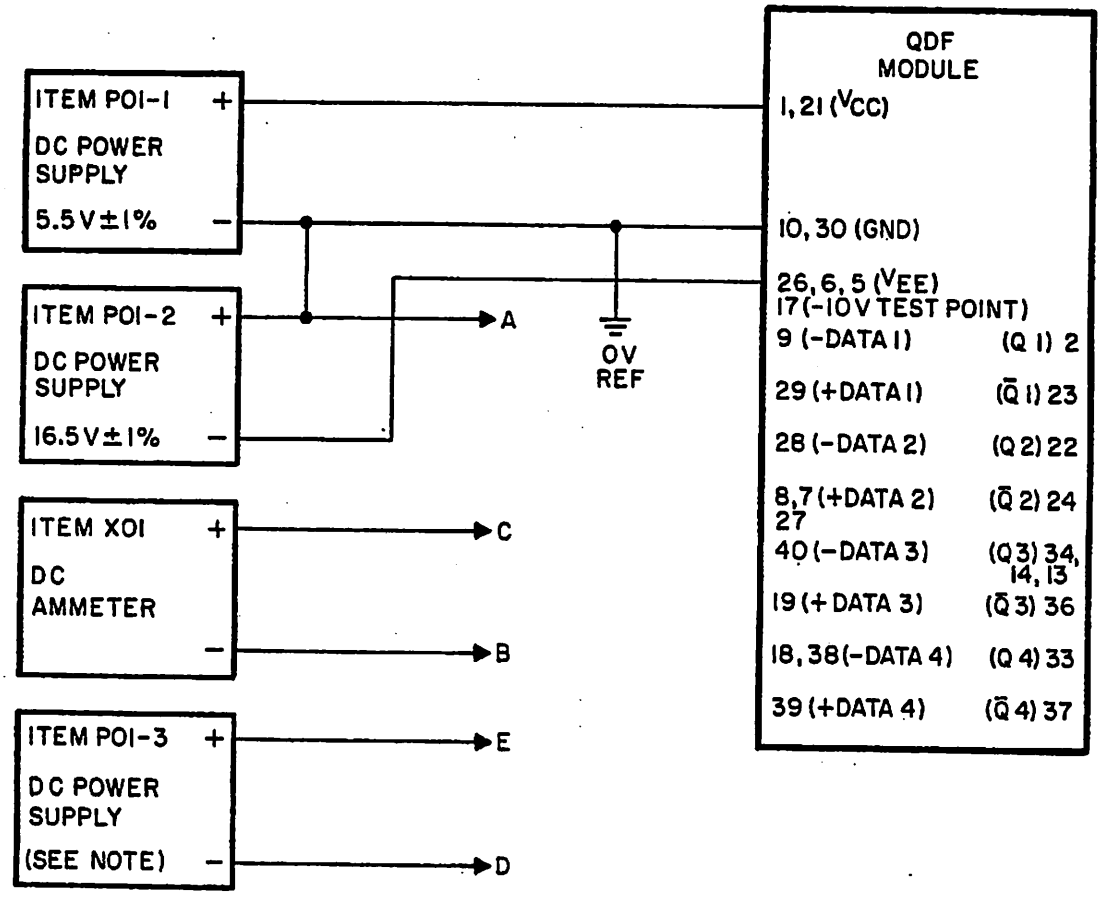
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
L1	Input current (V_{IN} = +10 V)	D,9	29	E	A,9	C	0.1	3.5	mA	13
L2		D,29	9		A,29					
L3		D,28	8,7,27		A,28					
L4		D,8,7,27	28		A,8,7,27					
L5		D,40	19		A,40					
L6		D,19	40		A,19					
L7		D,18,38	39		A,18,38					
L8		D,39	18,38		A,39					
M1	Input current (V_{IN} = -10 V)	E,9	29	D	C	A,9	-1.0	-7.0		
M2		E,29	9			A,29				
M3		E,28	8,7,27			A,28				
M4		E,8,7,27	28			A,8,7,27				
M5		E,40	19			A,40				
M6		E,19	40			A,19				
M7		E,18,38	39	C		A,18,38				
M8		E,39	18,38	D		A,39				

TABLE XIII. Input current (3 V and 0 V) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N1	Input current (V_{IN} = 3 V)	D,9	E	29	A,9	B	-1.0	+2.5	mA	13
N2		D,28		8,7,27	A,28					
N3		D,40		19	A,40					
N4		D,18,38		39	A;18,38					
N5	Input current (V_{IN} = 0 V)	C,9	29	A,9	Open	Open	-0.1	-2.5		
N6		C,28	8,7,27	A,28						

N7-8

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* NOTES: Set item POI-3 follows:
Test conditions L1-L8 = 10 V ±1%.
Test conditions M1-M8 = -10 V ±1%.
Test conditions N1-N4 = 3 V ±1%.
Test conditions N5-N8 = 0 V ±1%.

CAUTION:
Do not isolate the chassis of ammeter XOI from earth ground.

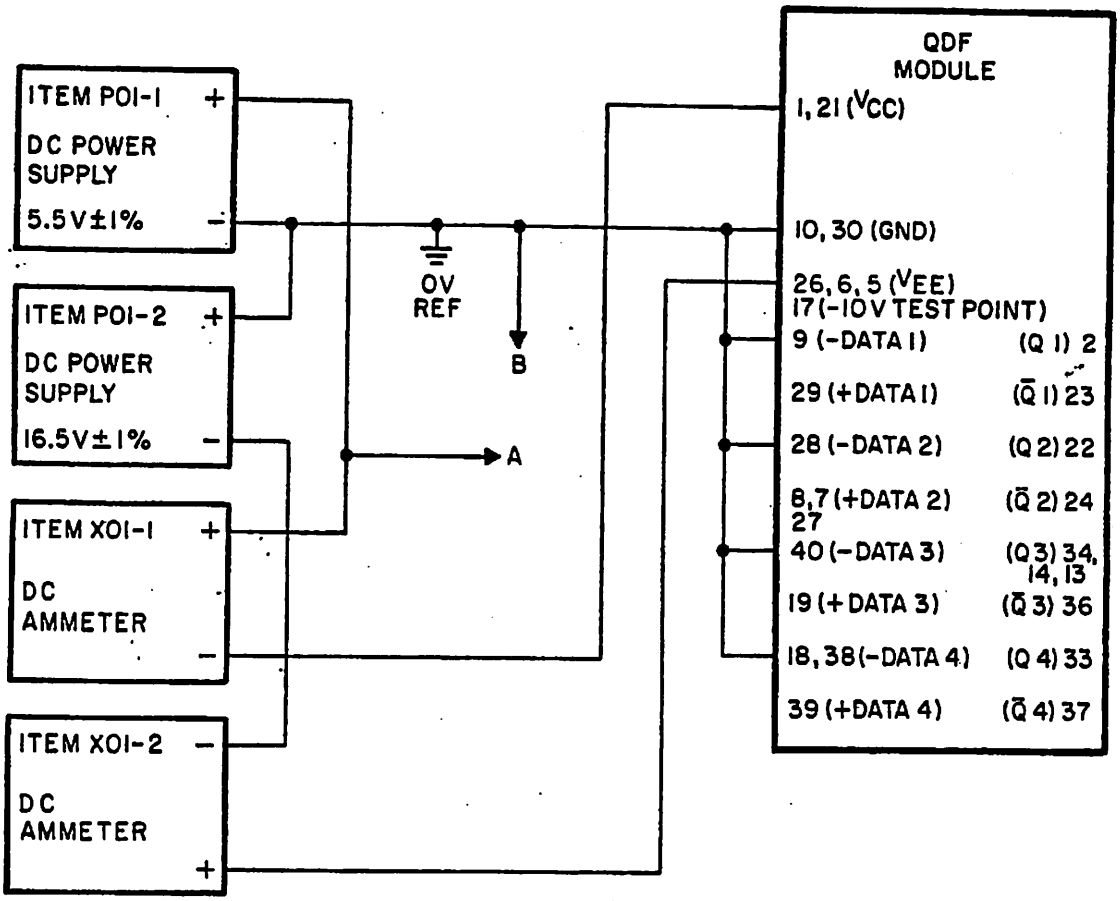
FIGURE 13. Input current (+10 V and -10 V differential inputs and 3 V and 0 V) test circuit for tables XII and XIII.

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TABLE XIV. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Power supply current, $\overline{Q}$ outputs low, Q outputs high.	Open	7,8,19,27, 29,39	10	49	mA	14
P2	Power supply current, $\overline{Q}$ outputs high, Q outputs low.	7,8,19,27, 29,39	Open	10	38		
Q1	Power supply current, $\overline{Q}$ outputs low, Q outputs high.	Open	7,8,19,27, 29,39	-15	-65		
Q2	Power supply current, $\overline{Q}$ outputs high, Q outputs low.	7,8,19,27, 29,39	Open	-15	-65		

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CAUTION:
Do not isolate the chassis of ammeters XOI-1 or XOI-2 from earth ground.

* FIGURE 14. Power supply current test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
R1	Q1 Transition time (t_{TLH}), C_L = 50 pF	29	2	9	10	300	ns	15,16
R2	Q1	9	2	29				
R3	Q2	8,7,27	22	28				
R4	Q2	28	22	8,7,27				
R5	Q3	19	34,14,13	40				
R6	Q3	40	34,14,13	19				
R7	Q4	39	33	18,38				
R8	Q4	18,38	33	39				
R9	$\overline{Q}$1	29	23	9	5	35		
R10	$\overline{Q}$1	9	23	29				
R11	$\overline{Q}$2	8,7,27	24	28				
R12	$\overline{Q}$2	28	24	8,7,27				
R13	$\overline{Q}$3	19	36	40				
R14	$\overline{Q}$3	40	36	19				
R15	$\overline{Q}$4	39	37	18,38				
R16	$\overline{Q}$4	18,38	37	39				
R17	Q1 Transition time (t_{THL}), C_L = 50 pF	29	2	9		30		
R18	Q1	9	2	29				
R19	Q2	8,7,27	22	28				
R20	Q2	28	22	8,7,27				
R21	Q3	19	34,14,13	40				
R22	Q3	40	34,14,13	19				
R23	Q4	39	33	18,38				
R24	Q4	18,38	33	39				
R25	$\overline{Q}$1	29	23	9		25		
R26	$\overline{Q}$1	9	23	29				
R27	$\overline{Q}$2	8,7,27	24	28				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R28	$\overline{Q}2$ Transition time (t_{THL}), C_L = 50 pF	28	24	8,7,27	~~10~~ 5	25	ns	15,16
R29	$\overline{Q}3$	19	36	40				
R30	$\overline{Q}3$	40	36	19				
R31	$\overline{Q}4$	39	37	18,38				
R32	$\overline{Q}4$	18,38	37	39				
R33	Q1 Transition time (t_{TLH}), C_L = 150 pF	29	2	9	10	310		
R34	Q1	9	2	29				
R35	Q2	8,7,27	22	28				
R36	Q2	28	22	8,7,27				
R37	Q3	19	34,14,13	40				
R38	Q3	40	34,14,13	19				
R39	Q4	39	33	18,38				
R40	Q4	18,38	33	39				
R41	$\overline{Q}1$	29	23	9	5	40		
R42	$\overline{Q}1$	9	23	29				
R43	$\overline{Q}2$	8,7,27	24	28				
R44	$\overline{Q}2$	28	24	8,7,27				
R45	$\overline{Q}3$	19	36	40				
R46	$\overline{Q}3$	40	36	19				
R47	$\overline{Q}4$	39	37	18,38				
R48	$\overline{Q}4$	18,38	37	39				
R49	Q1 Transition time (t_{THL}), C_L = 150 pF	29	2	9		35		
R50	Q1	9	2	29				
R51	Q2	8,7,27	22	28				
R52	Q2	28	22	8,7,27				
R53	Q3	19	34,14,13	40				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R54	Q3 Transition time (t_{THL}), C_L = 150 pF	40	34,14,13	19	~~10~~ 5	35	ns	15,16
R55	Q4	39	33	18,38				
R56	Q4	18,38	33	39				
R57	$\overline{Q}$1	29	23	9		30		
R58	$\overline{Q}$1	9	23	29				
R59	$\overline{Q}$2	8,7,27	24	28				
R60	$\overline{Q}$2	28	24	8,7,27				
R61	$\overline{Q}$3	19	36	40				
R62	$\overline{Q}$3	40	36	19				
R63	$\overline{Q}$4	39	37	18,38				
R64	$\overline{Q}$4	18,38	37	39				

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	Q1 Propagation delay time (t_{PLH}), C_L = 50 pF	29	2	9	5	52	ns	15,17
S2	Q1	9	2	29				
S3	Q2	8,7,27	22	28				
S4	Q2	28	22	8,7,27				
S5	Q3	19	34,14,13	40				
S6	Q3	40	34,14,13	19				
S7	Q4	39	33	18,38				
S8	Q4	18,38	33	39				
S9	$\overline{Q1}$	29	23	9	8	76		
S10	$\overline{Q1}$	9	23	29				
S11	$\overline{Q2}$	8,7,27	24	28				
S12	$\overline{Q2}$	28	24	8,7,27				
S13	$\overline{Q3}$	19	36	40				
S14	$\overline{Q3}$	40	36	19				
S15	$\overline{Q4}$	39	37	18,38				
S16	$\overline{Q4}$	18,38	37	39				
S17	Q1 Propagation delay time (t_{PHL}), C_L = 50 pF	29	2	9	5	52		
S18	Q1	9	2	29				
S19	Q2	8,7,27	22	28				
S20	Q2	28	22	8,7,27				
S21	Q3	19	34,14,13	40				
S22	Q3	40	34,14,13	19				
S23	Q4	39	33	18,38				
S24	Q4	18,38	33	39				
S25	$\overline{Q1}$	29	23	9	8	76		
S26	$\overline{Q1}$	9	23	29	8	76		

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

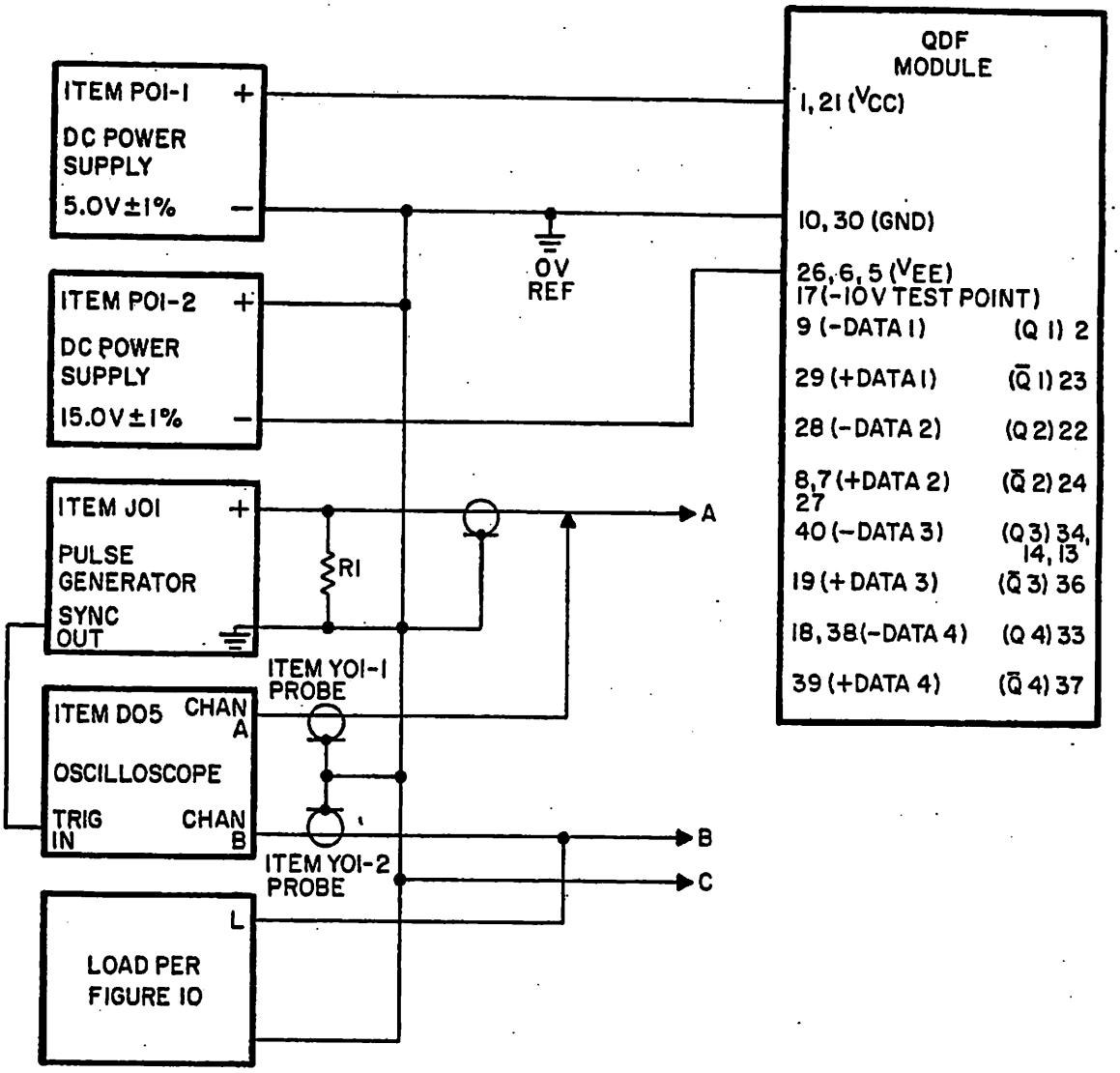
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S27	$\overline{Q}2$ Propagation delay time (t_{PHL}), C_L = 50 pF	8,7,27	24	28	8	76	ns	15,17
S28	$\overline{Q}2$	28	24	8,7,27				
S29	$\overline{Q}3$	19	36	40				
S30	$\overline{Q}3$	40	36	19				
S31	$\overline{Q}4$	39	37	18,38				
S32	$\overline{Q}4$	18,38	37	39				
S33	Q1 Propagation delay time (t_{PLH}), C_L = 150 pF	29	2	9	5	58		
S34	Q1	9	2	29				
S35	Q2	8,7,27	22	28				
S36	Q2	28	22	8,7,27				
S37	Q3	19	34,14,13	40				
S38	Q3	40	34,14,13	19				
S39	Q4	39	33	18,38				
S40	Q4	18,38	33	39				
S41	$\overline{Q}1$	29	23	9	8	82		
S42	$\overline{Q}1$	9	23	29				
S43	$\overline{Q}2$	8,7,27	24	28				
S44	$\overline{Q}2$	28	24	8,7,27				
S45	$\overline{Q}3$	19	36	40				
S46	$\overline{Q}3$	40	36	19				
S47	$\overline{Q}4$	39	37	18,38				
S48	$\overline{Q}4$	18,38	37	39				
S49	Q1 Propagation delay time (t_{PHL}), C_L = 150 pF	29	2	9	5	58		
S50	Q1	9	2	29				
S51	Q2	8,7,27	22	28				
S52	Q2	28	22	8,7,27				

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S53	Q3 Propagation delay time (t_{PHL}), C_L = 150 pF	19	34,14,13	40	5	58	ns	15,17
S54	Q3	40	34,14,13	19				
S55	Q4	39	33	18,38				
S56	Q4	18,38	33	39				
S57	$\overline{Q}$1	29	23	9	8	82		
S58	$\overline{Q}$1	9	23	29				
S59	$\overline{Q}$2	8,7,27	24	28				
S60	$\overline{Q}$2	28	24	8,7,27				
S61	$\overline{Q}$3	19	36	40				
S62	$\overline{Q}$3	40	36	19				
S63	$\overline{Q}$4	39	37	18,38				
S64	$\overline{Q}$4	18,38	37	39				

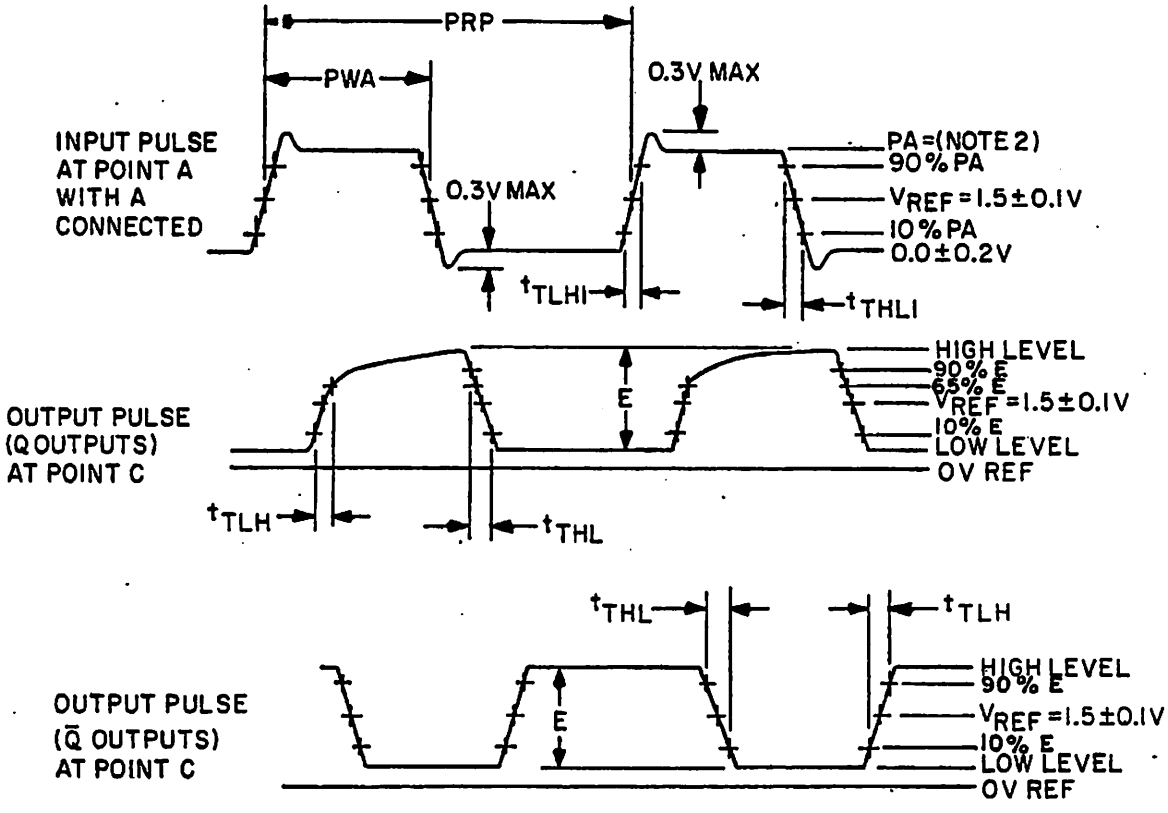
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV and XVI.

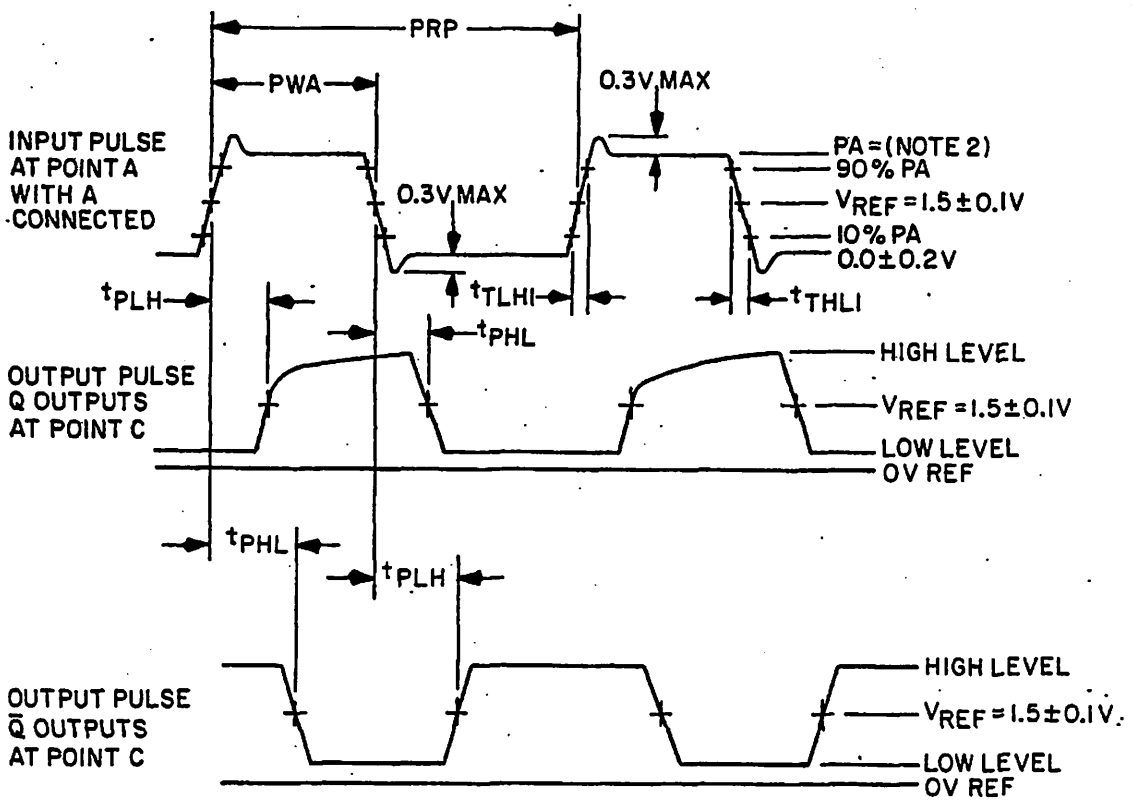
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. For odd numbered tests (R1 through R64) PA = +3.0 ±0.1 V. For even numbered tests (R1 through R64) PA = -3.0 ±0.1 V.
3. For odd numbered test conditions R, $\overline{Q}$ output is in phase with respect to the input and Q output is out of phase with respect to the input. For even numbered test conditions R, $\overline{Q}$ output is out of phase with respect to the input and Q output is in phase with respect to the input.
4. Before making the measurements specified in table XV , verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
5. The lead length of electrical conductors between resistor R1, and input pins, probes Y01-1 and Y01-2 to output pins and between load circuits and output pins shall be less than 2 inches (50.8 mm).
6. All voltages referenced to ground.

* FIGURE 16. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XV.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. For odd numbered tests (S1 through S64) PA = +3.0 ±0.1 V. For even numbered tests (S1 through S64) PA = -3.0 ±0.1 V.
3. For odd numbered test conditions S, $\overline{Q}$ output is in phase with respect to the input and Q output is out of phase with respect to the input. For even numbered test conditions S, $\overline{Q}$ output is out of phase with respect to the input and Q output is in phase with respect to the input.
4. Before making the measurements specified in tables XV , verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
5. The lead length of electrical conductors between resistor R1, and input pins, probes YO1-1 and YO1-2 to output pins and between load circuits and output pins shall be less than 2 inches (50.8 mm).
6. All voltages referenced to ground.

* FIGURE 17. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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TABLE XVII. Common mode rejection check (low-level and high-level) test.

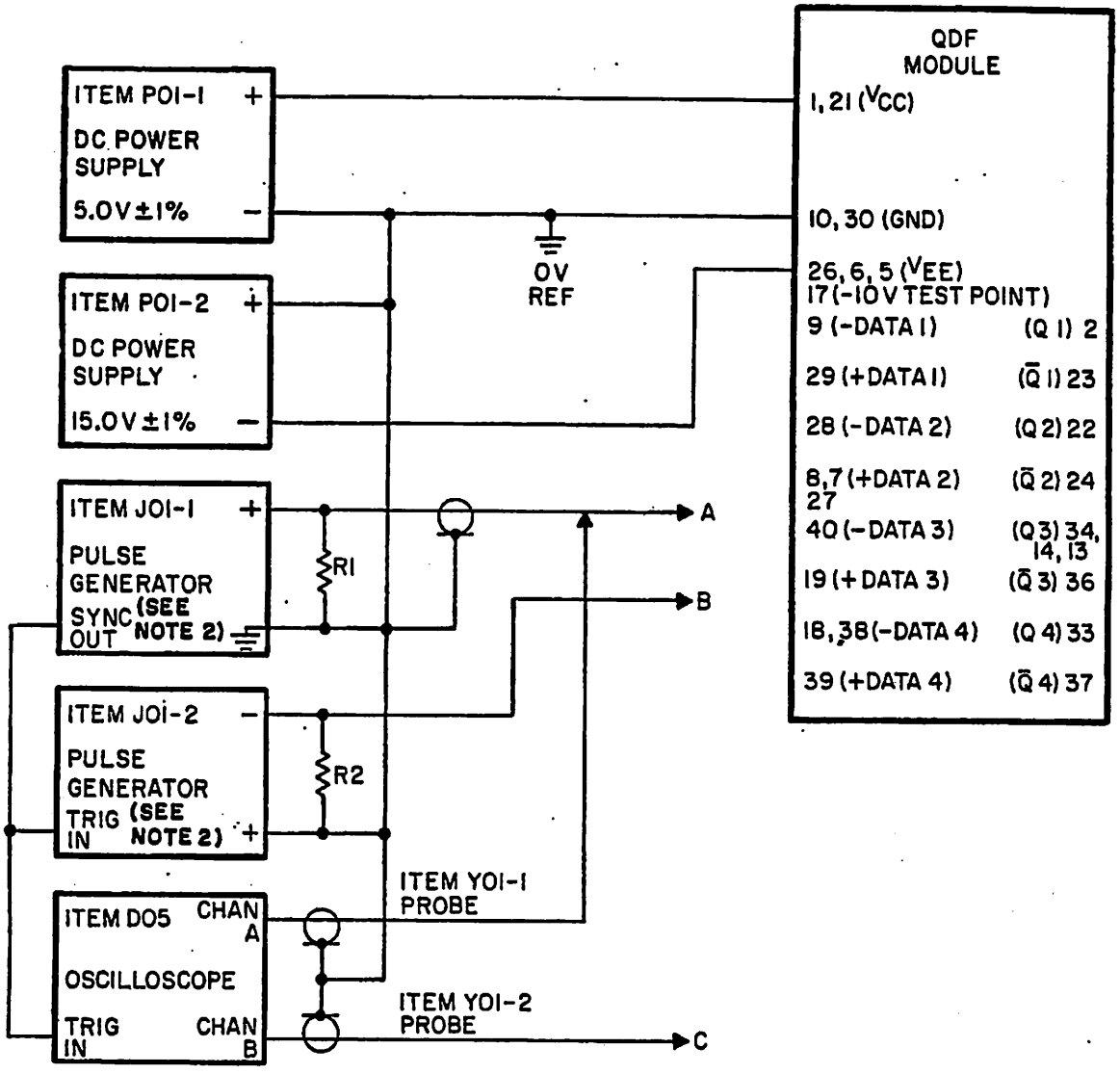
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	Q1 Common mode rejection check, low level	9	29	2	0.0	0.4	V	18,19
T2	Q2	28	8,7,27	22				
T3	Q3	40	19	34,14,13				
T4	Q4	18,38	39	33				
T5	$\overline{Q}$1	9	29	23				
T6	$\overline{Q}$2	28	8,7,27	24				
T7	$\overline{Q}$3	40	19	36				
T8	$\overline{Q}$4	18,38	39	37				
T9	Q1	9	29	2				
T10	Q2	28	8,7,27	22				
T11	Q3	40	19	34,14,13				
T12	Q4	18,38	39	33				
T13	$\overline{Q}$1	9	29	23				
T14	$\overline{Q}$2	28	8,7,27	24				
T15	$\overline{Q}$3	40	19	36				
T16	$\overline{Q}$4	18,38	39	37				
U1	Q1 Common mode rejection check, high level	9	29	2	2.8	4.5		
U2	Q2	28	8,7,27	22				
U3	Q3	40	19	34,14,13				
U4	Q4	18,38	39	33				
U5	$\overline{Q}$1	9	29	23	2.4			
U6	$\overline{Q}$2	28	8,7,27	24				
U7	$\overline{Q}$3	40	19	36				
U8	$\overline{Q}$4	18,38	39	37				
U9	Q1	9	29	2	2.8			
U10	Q2	28	8,7,27	22	2.8			

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TABLE XVII. Common mode rejection check (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U11	Q3 Common mode rejection check, high level	40	19	34,14,13	2.8	4.5	V	18,19
U12	Q4	18,38	39	33	2.8			
U13	$\overline{Q1}$	9	29	23	2.4			
U14	$\overline{Q2}$	28	8,7,27	24				
U15	$\overline{Q3}$	40	19	36				
U16	$\overline{Q4}$	18,38	39	37				

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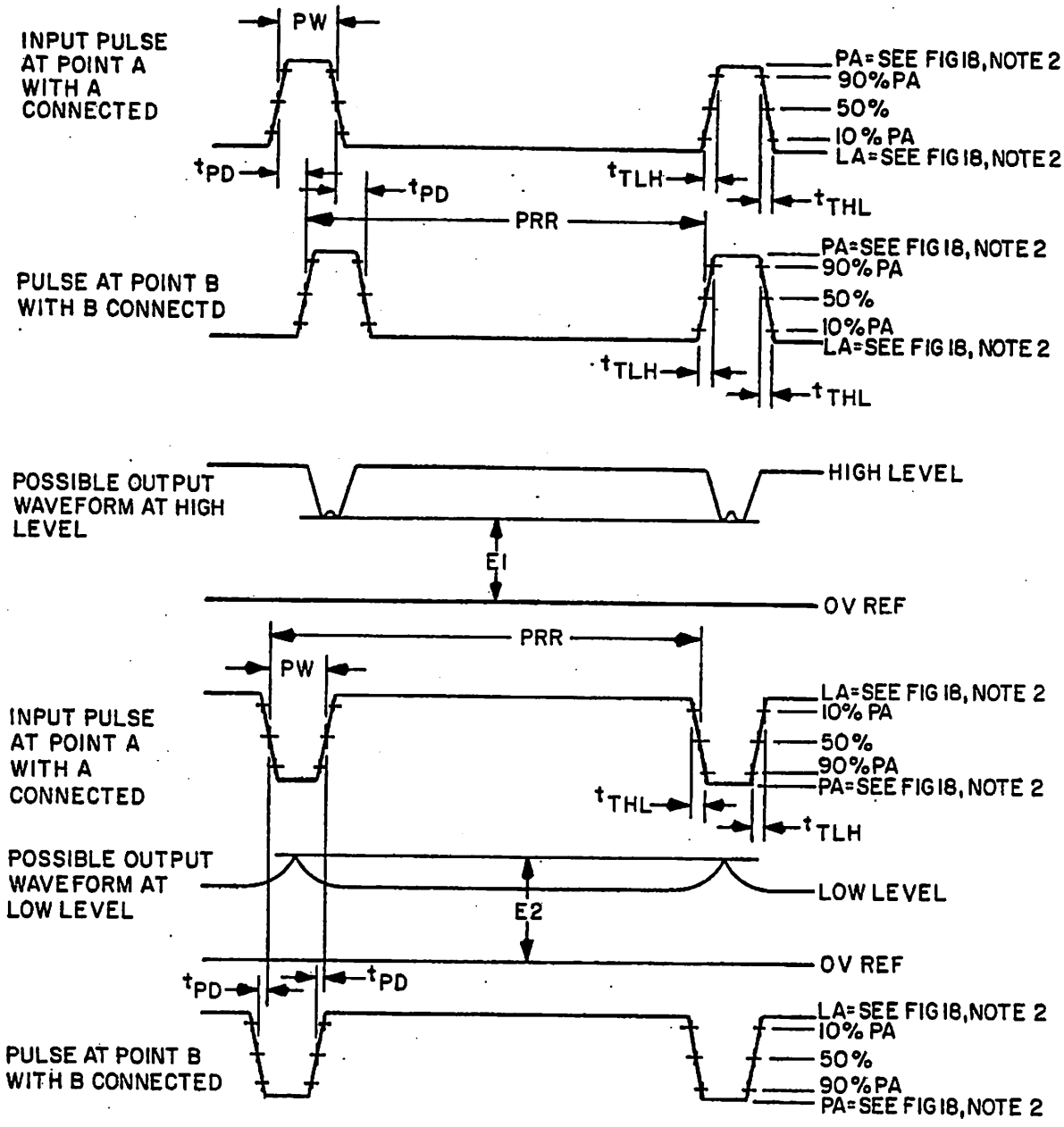
NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Input pulse amplitude parameters:

Test condition	Item J01-1	Item J01-2
T1-T4, U5-U8	PA = 6.0 ±0.1 V, LA = 0.0 ±0.1 V	PA = 8.2 ±0.1 V, LA = 2.2 ±0.1 V
T5-T8, U1-U4	PA = 6.0 ±0.1 V, LA = 0.0 ±0.1 V	PA = 6.8 ±0.1 V, LA = 0.8 ±0.1 V
T9-T12, U13-U16	PA = -6.0 ±0.1 V, LA = 0.0 ±0.1 V	PA = -3.8 ±0.1 V, LA = 2.2 ±0.1 V
T13-T16, U9-U12	PA = -6.0 ±0.1 V, LA = 0.0 ±0.1 V	PA = -5.2 ±0.1 V, LA = 0.8 ±0.1 V

3. Load output under test with load specified on figure 10.

* FIGURE 18. Common mode rejection check (low-level and high-level) test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PW = 50 ±2 ns.
* PRP = 200 ±20 ns.
   $t_{TLH} = t_{THL} = 5 \pm 1$ ns.
   $t_{PD} = 0 \pm 1.0$ ns.
2. E1 is measured from 0 V REF to the most negative peak of the high level waveform.
3. E2 is measured from 0 V REF to the most positive peak of the low level waveform.

FIGURE 19. Common mode rejection check (low-level and high-level) test circuit waveforms for table XVII.

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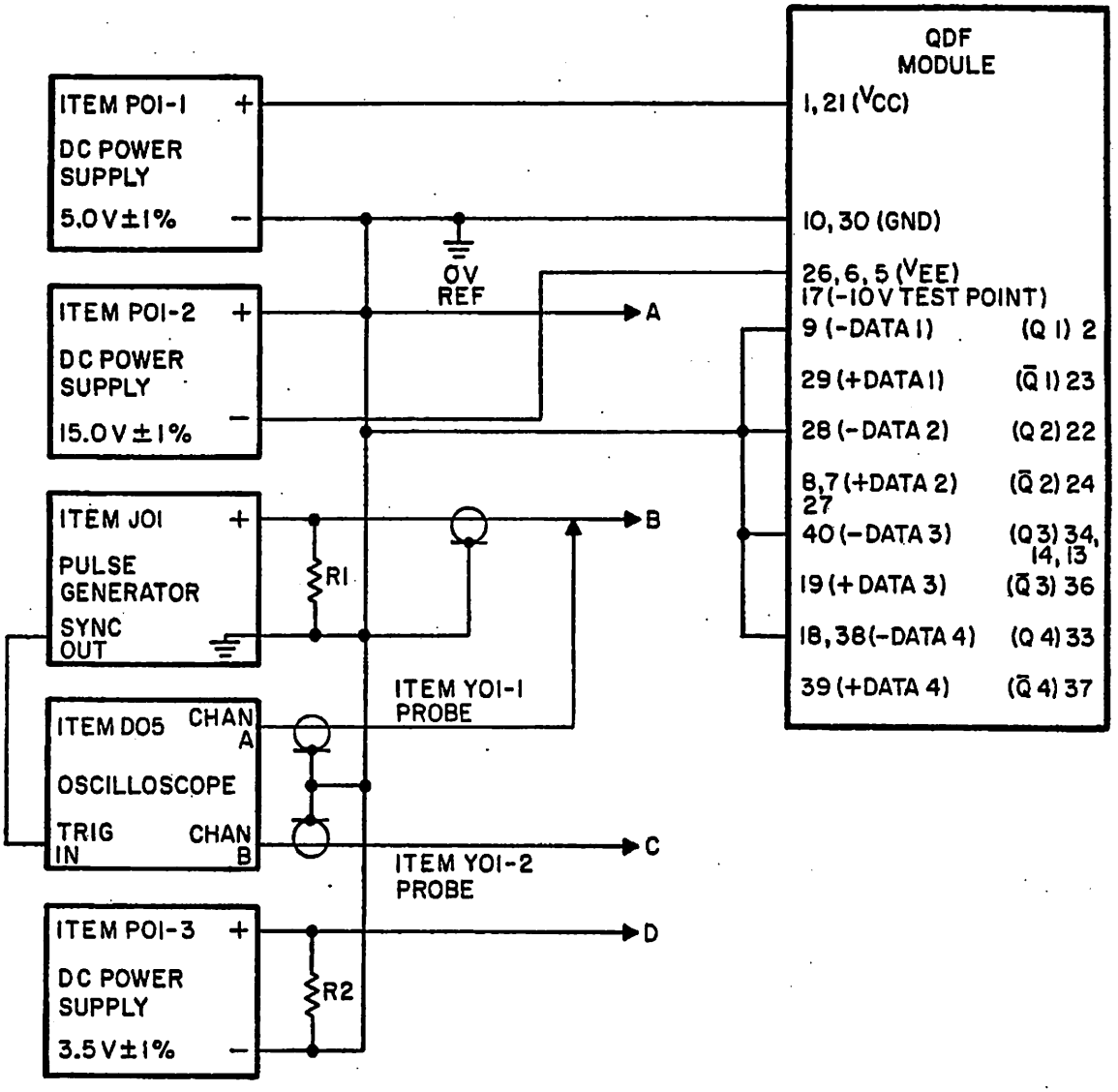
*TABLE XVIII. Output crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	Q1 Output crosstalk, low level	Open	8,7,27,19, 39	2	29	0	400	mV	20,21
V2	$\overline{Q1}$	29	8,7,27,19, 39	23	Open				
V3	Q2	Open	29,19,39	22	8,7,27				
V4	$\overline{Q2}$	8,7,27	29,19,39	24	Open				
V5	Q3	Open	29,8,7,27, 39	34,14,13	19				
V6	$\overline{Q3}$	19	29,8,7,27, 39	36	Open				
V7	Q4	Open	29,8,7,27, 19	33	39				
V8	$\overline{Q4}$	39	29,8,7,27, 19	37	Open				

*TABLE XIX. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	Q1 Output crosstalk, high level	29	8,7,27,19, 39	2	Open	0	400	mV	20,21
W2	$\overline{Q1}$	Open	8,7,27,19, 39	23	29				
W3	Q2	8,7,27	29,19,39	22	Open				
W4	$\overline{Q2}$	Open	29,19,39	24	8,7,27				
W5	Q3	19	29,8,7,27, 39	34,14,13	Open				
W6	$\overline{Q3}$	Open	29,8,7,27, 39	36	19				
W7	Q4	39	29,8,7,27, 19	33	Open				
W8	$\overline{Q4}$	Open	29,8,7,27, 19	37	39				

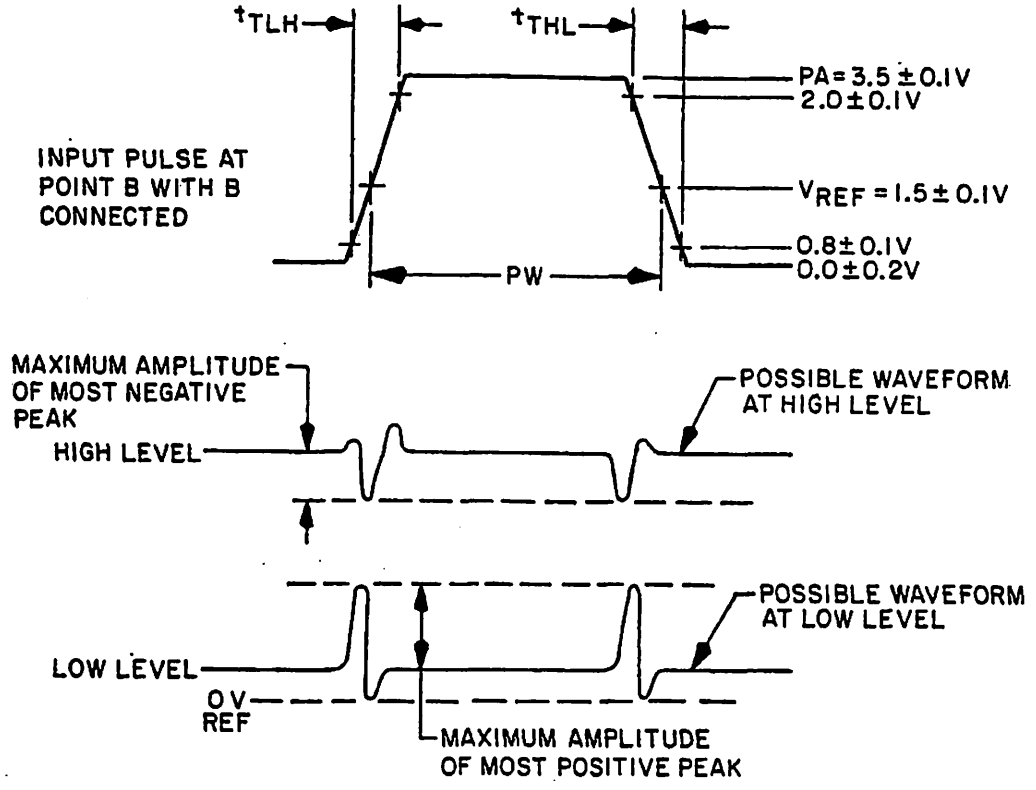
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load all outputs with load as specified on figure 10.

FIGURE 20. Output crosstalk (low-level and high-level) test circuit for tables XVIII and XIX.

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NOTE: Input pulse parameters:
PW = 200 ns ±10%.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PRR = 1 MHz ±10%.

FIGURE 21. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVIII and XIX.

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TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
X	Power supply transient amplitude	0	400	mV	22,23

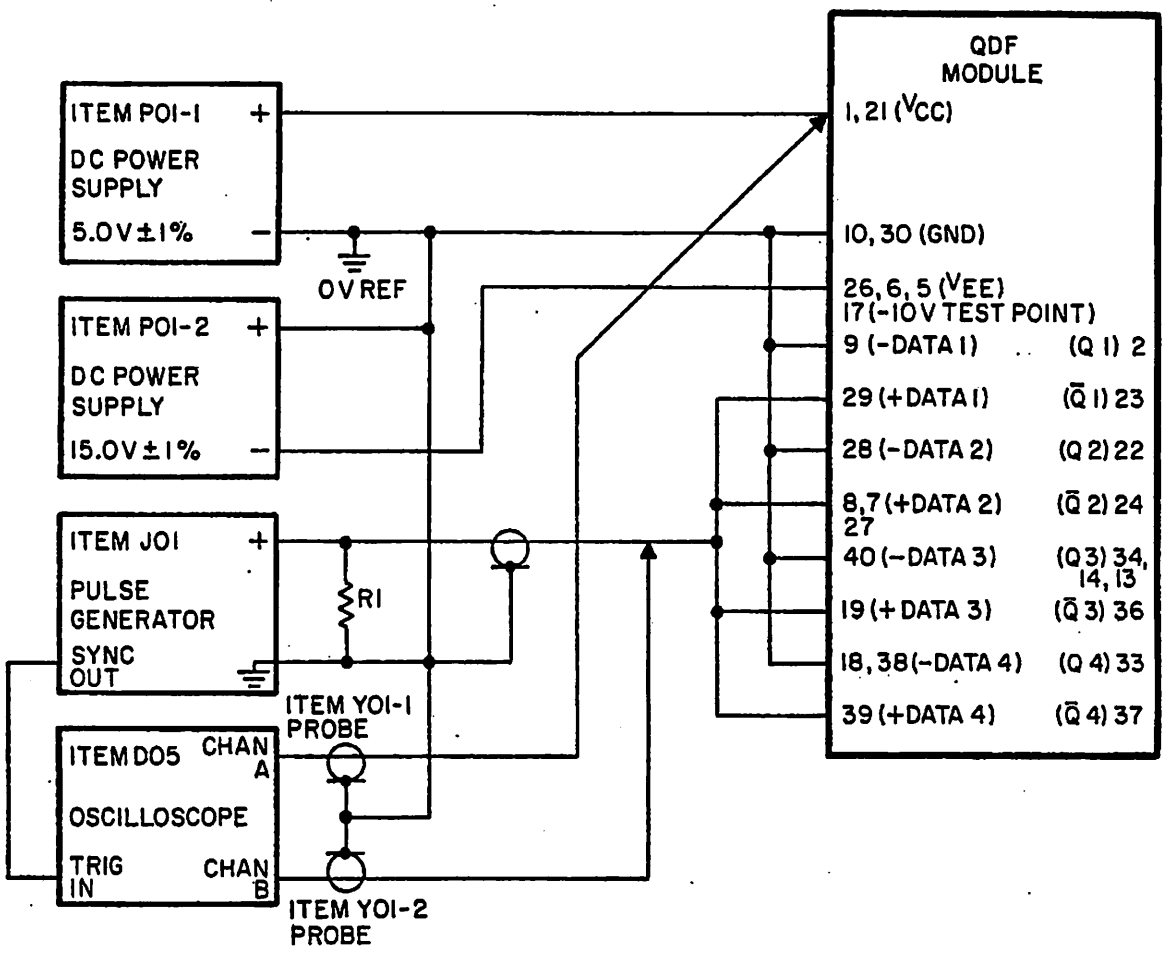
* TABLE XXI. Power supply transient rise time test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Y	Power supply transient rise time	1	3.0	8.5	µs	24,25

TABLE XXII. Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Z1	Input capacitance	9,29	5	35	pF	26
Z2	↓	28,8,7,27	↓	↓	↓	↓
Z3		19,40				
Z4		18,38,39				

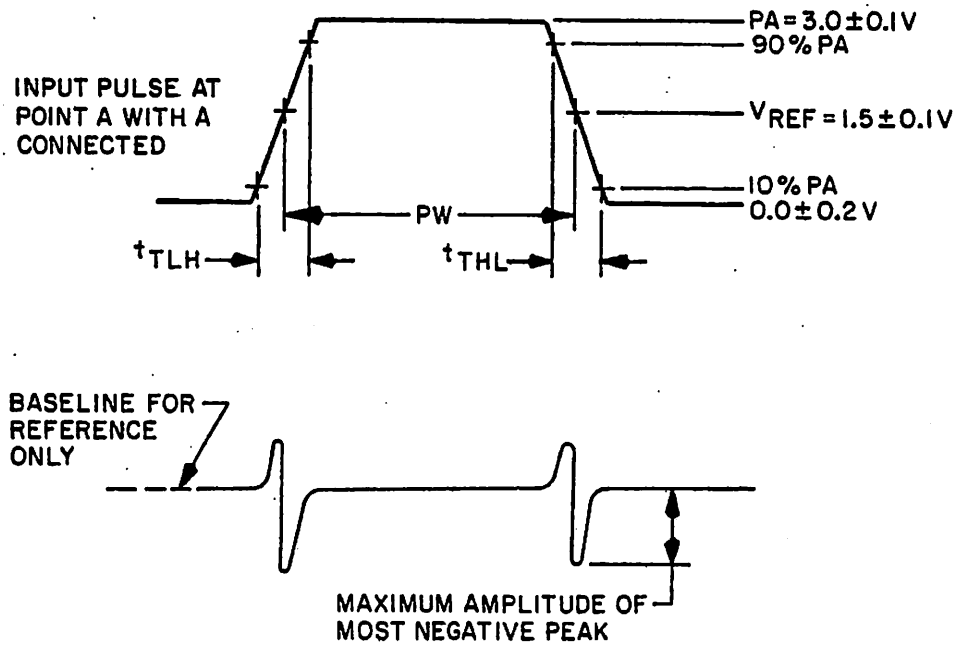
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 22. Power supply transient amplitude test circuit for table XX.

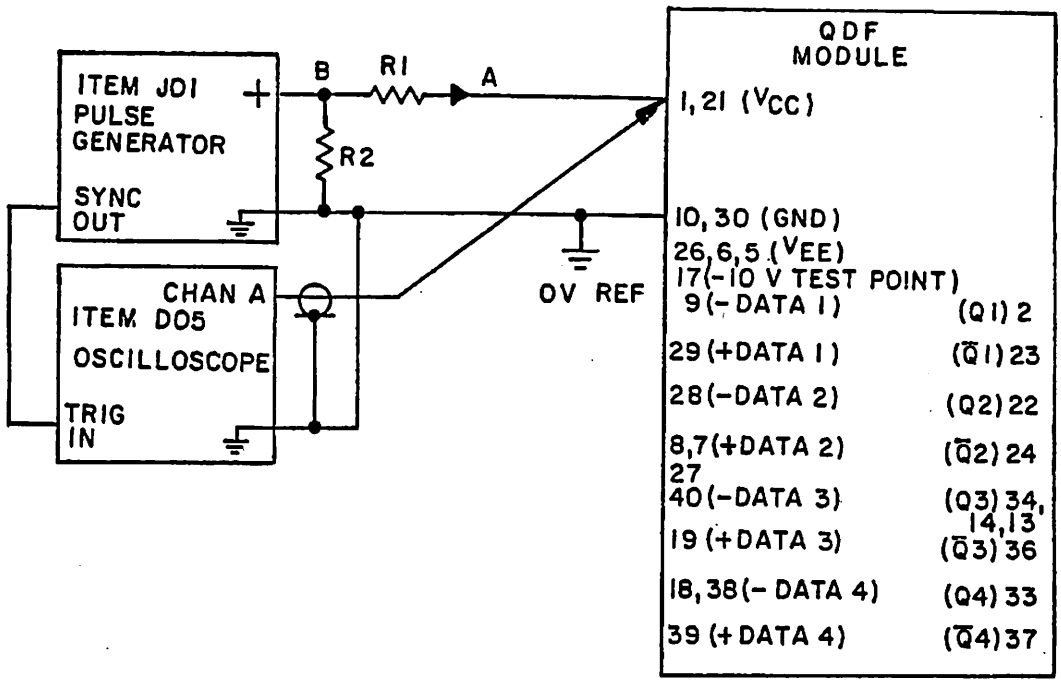
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NOTE: Input pulse parameters:
PW = 500 ns ±10%.
$t_{TLH} = t_{THL}$ = 10 ±1 ns.
PRR = 1 MHz ±10%.

FIGURE 23. Power supply transient amplitude test circuit waveforms for table XX.

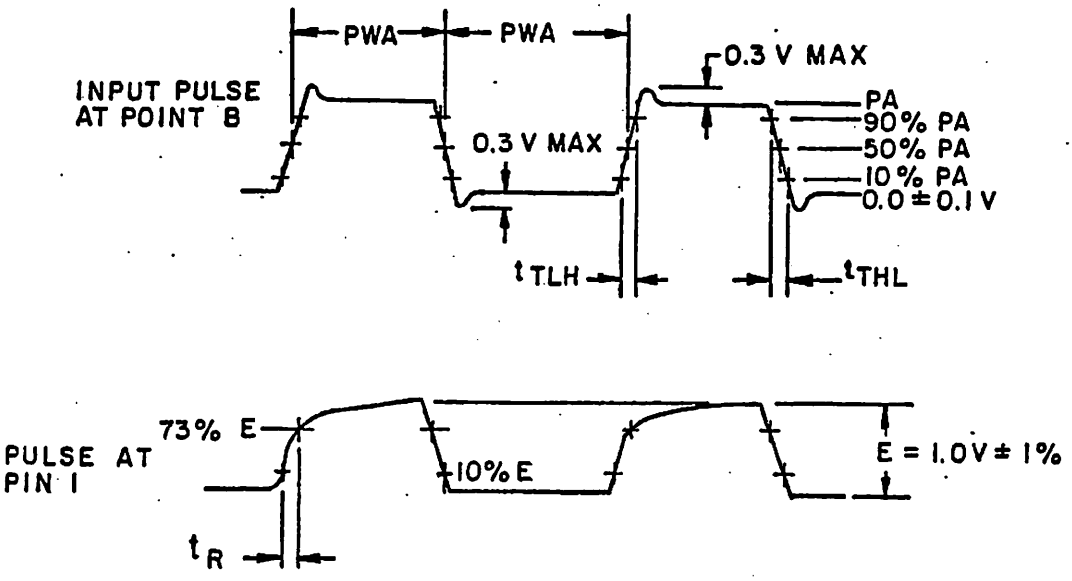
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NOTES:
1. R1 = 511Ω ±1%, 1/4 W.
2. R2 = 51.1Ω ±5%, 1/2 W.

* FIGURE 24. Power supply transient rise time test circuit for table XXI.

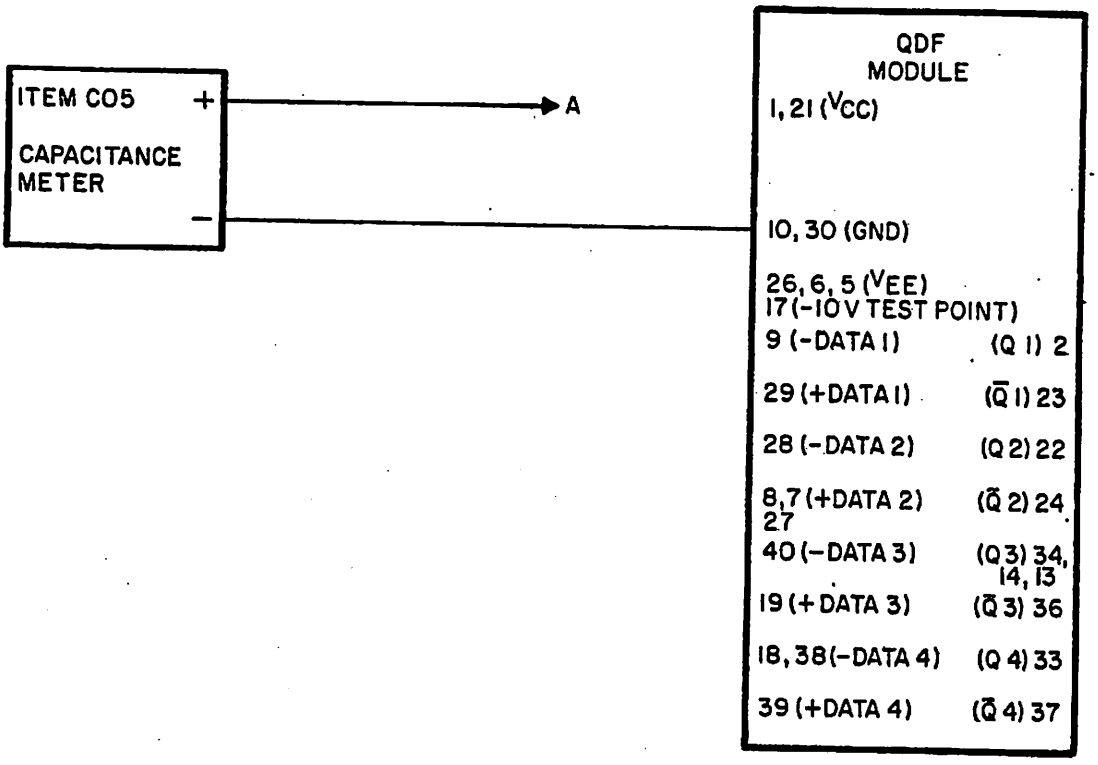
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NOTES:
1. Input pulse parameters:
   PA = 1.0 V ±1%.
   PWA = 40 ±1 µs.
   $t_{TLH} = t_{THL} = 20 \pm 2$ ns.
2. t_R is measured from the 10% to 73% levels.

*FIGURE 25. Power supply transient rise time test circuit waveforms for table XXI.

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NOTES:
1. Null out the effects of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pins under test.

FIGURE 26. Input capacitance test circuit for table XXII.

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